

[SCHOOL NAME — PLACEHOLDER LETTERHEAD]

(Swap in your school's actual letterhead / logo here)

Class 4 Maths — CA-1 (Hard) — Answer Key

Class : 4

Subject : Maths (Math-Mela)

Max Marks : 30

Duration : 1 Hour

Syllabus: Chapters 1–3 — Shapes Around Us, Hide and Seek, Pattern Around Us

General Instructions: All questions are compulsory. For Section A, write the correct option letter.

ANSWER KEY

SECTION A — Multiple Choice Questions (1 Mark each)

1. A solid shape has 5 faces, 9 edges, and 6 corners. Using Faces + Corners - Edges = 2, verify if this could be a valid solid, and identify it. (1 Mark)

A. $5 + 6 - 9 = 2$, valid; likely a triangular prism ✓

B. $5 + 6 - 9 = 0$, invalid

C. $5 + 9 - 6 = 8$, invalid

D. The formula cannot be used here

2. If you see only the top view and front view of an object but not the side view, what limitation might you face in fully understanding its shape? (1 Mark)

A. No limitation, two views are always enough

B. You might not know the exact depth or shape variations only visible from the side ✓

C. Top and front views are identical, so no information is lost

D. It becomes impossible to know anything about the object

3. A sequence of numbers goes 2, 4, 6, 8, What is the 10th number in this pattern? (1 Mark)

A. 18

B. 20 ✓

C. 22

D. 16

4. Which of these combinations of coins/notes to make ₹145 uses the FEWEST number of pieces (available: 100, 50, 20, 10, 5)? (1 Mark)

A. $100 + 20 + 20 + 5$ ✓

B. $50 + 50 + 20 + 20 + 5$

C. $100 + 50 - 5$ (not valid)

D. $20 \times 7 + 5$

5. A pyramid has a square base. How many triangular faces will it have in addition to its base? (1 Mark)

A. 2

B. 3

C. 4 ✓

D. 5

6. If a grid position is described as "2 rows down and 3 columns right" from the top-left corner, which of these best describes this position? (1 Mark)

A. Row 2, Column 3 ✓

B. Row 3, Column 2

C. Row 1, Column 1

D. It cannot be determined

7. Which statement is TRUE about the sum of two odd numbers? (1 Mark)

A. The sum is always odd

B. The sum is always even ✓

C. The sum can be either odd or even

D. Odd numbers cannot be added together

SECTION B — Short Answer Questions (2 Marks each)

1. Explain why knowing only the top view of a building is not enough to fully construct it, using reasoning about missing information. (0 Marks)

► The top view only shows the layout and outline of the building from above, but it does not show the height of walls, the design of the front entrance, or other vertical details, all of which are necessary for actually constructing the building, so front and side views are also needed.

2. A pattern of numbers goes 3, 7, 11, 15, Identify the rule of the pattern and find the next two numbers. (0 Marks)

► The rule is adding 4 to the previous number each time ($3+4=7$, $7+4=11$, $11+4=15$); following this rule, the next two numbers are 19 and 23.

3. Explain why a triangular pyramid (with a triangular base) has fewer faces than a square pyramid (with a square base), using their properties. (0 Marks)

► A triangular pyramid has 1 triangular base + 3 triangular side faces = 4 faces total, while a square pyramid has 1 square base + 4 triangular side faces = 5 faces total; the difference comes from the base shape needing more side faces to enclose a square base than a triangular one.

4. A group of 47 students is being divided for two different games - one needing pairs, and one needing groups of 5. Explain which grouping will have students left over and why. (0 Marks)

► For pairs (groups of 2), 47 is odd, so 23 pairs can be formed with 1 student left over; for groups of 5, $47 \div 5 = 9$ remainder 2, so 9 groups of 5 can be formed with 2 students left over - both groupings leave some students without a complete group.

5. Explain how understanding both grid positions and directions (like up, down, left, right) together helps in describing movement, such as in a simple maze. (0 Marks)

► Grid positions tell you exactly where something is located (like row 3, column 2), while directions tell you how to move from one position to another (like "move right two columns"); together, they allow precise navigation instructions, such as solving or describing a path through a maze.

6. A cube net (unfolded cube) is shown with 6 squares arranged in a cross shape. Explain how you would verify that this net can actually fold into a cube. (0 Marks)

- To verify, you check that there are exactly 6 square faces (matching a cube's 6 faces), and that when mentally folded along the edges, each square would align to form a closed 3D shape with no overlaps or gaps, matching all of a cube's properties.

SECTION C — Short Answer Questions (Explain) (3 Marks each)

1. Analyse how the properties of 3D shapes (faces, edges, corners) follow a predictable mathematical relationship ($\text{Faces} + \text{Corners} - \text{Edges} = 2$), and discuss why this rule is useful for verifying whether a described shape is possible. (0 Marks)

- This rule, known as Euler's formula, shows that the numbers of faces, corners, and edges of many solid shapes are not random but follow a consistent mathematical relationship.
- By applying this formula to a described shape, we can check if the given numbers are mathematically possible - for example, a shape with 5 faces, 6 corners, and 9 edges checks out ($5+6-9=2$), confirming it could be a valid solid like a triangular prism.
- This is useful because it allows us to verify shape properties logically, rather than only through physical observation, which is especially helpful when working with descriptions or diagrams rather than real 3D objects.

2. Discuss how combining knowledge of views (top, front, side) with grid-based positioning creates a more complete spatial understanding than either skill alone. (0 Marks)

- Understanding different views (top, front, side) helps us understand the shape and structure of an object from multiple angles, which is important for recognising 3D objects fully.
- Grid-based positioning helps us describe exact locations within a space, which is useful for navigation and organisation, such as finding a specific desk in a classroom.
- Combining both skills allows for a genuinely complete spatial understanding - being able to both recognise what something looks like from different angles AND pinpoint exactly where it is located, which is valuable in fields like architecture, mapping, and even video game design.

SECTION D — Long Answer Questions (5 Marks each)

1. Write a detailed essay analysing how the three chapters studied (Shapes Around Us, Hide and Seek, Pattern Around Us) connect to real-world careers or fields, such as architecture, design, or data analysis, explaining the specific skills from each chapter that would be useful. (0 Marks)

- "Shapes Around Us" directly connects to careers like architecture and engineering, where understanding the properties of 2D and 3D shapes (faces, edges, corners, angles) is essential for designing stable, functional structures.
- "Hide and Seek" connects to fields like architecture, urban planning, and even game design, where understanding different views (top, front, side) and spatial positioning (grids, directions) is crucial for creating accurate plans, maps, or virtual environments.
- "Pattern Around Us" connects to fields like data analysis, computer programming, and finance, where recognising numerical patterns, understanding even/odd classifications, and efficiently managing quantities (like money) are foundational skills.
- Together, these three chapters, though taught at a basic level in Class 4, lay the groundwork for skills that professionals in many different fields continue to build upon and use throughout their careers, showing the real-world relevance of early mathematics education.